

Exercise 2 Management Summary

Group A - DOPP

January 8, 2026

1 Introduction

The EU's waste policy under the Circular Economy sets recycling targets for municipal, packaging, and e-waste, backed by directives. While the EU weighted average has risen to roughly half of municipal waste recycled, large country disparities persist. This project uses data published by Eurostat, Weltbank and Our World in Data to analyze trends over time, compare countries and waste streams, explore potential drivers, and evaluate predictability of recycling developments. Five questions have been formulated for analysis: How has the amount of recycling of waste developed in the EU overall over time? How does recycling compare across countries in the EU? How does it compare for specific types of waste such as municipal waste, packaging waste, and electrical waste? Are there characteristics of countries that could lead to increased recycling? How well can the development of the amount of recycling be predicted?

2 Methodology

The analysis has been conducted following the CRISP-DM framework. To answer the first three questions, statistical tests and visual analytics techniques have been applied on an EU aggregate and individual country as well as waste stream level. Those countries showing major up- or downside developments during the observed period of time have been analyzed in more detail with the help of Gemini 3.5 deep research to link significant trends to policies or external factors that have potentially caused the respective movement. Question 4 is explored using panel regression analysis for within-country changes over time, controlling for unobserved, time-invariant factors, while PCA and Hierarchical Clustering are used for across-country analysis to identify common patterns and groupings across countries. For the predictions of the development of recycling, regression models (simple linear regression and panel regression) and time series models are used.

3 Findings

3.1 How has the amount of recycling of waste developed in the EU overall over time?

Our analysis of the EU27 aggregate from 2000 to 2023 reveals a fundamental structural shift. The EU has successfully transitioned from a **disposal-based** to a **recovery-based** waste management model. This is backed by the Kruskal & Spearman test which indicating significant changes of recycling rates over time.

As shown in Figure 1, while the absolute volume of treated waste remained relatively stable (growing slightly from 209 to 218 million tonnes), the composition of treatment changed drastically. In 2000, the majority of municipal waste (61.6%) was disposed of via landfills or incineration without energy. By 2023, this ratio has effectively flipped now **76.6% of waste**

is now recovered by Recycling and Energy Recovery. Further, the material recycling rate (green bars) climbed consistently from 28.8% in 2000 to reach a milestone of **50.0% in 2023**. In absolute terms, recycling volumes nearly doubled from around 60 million to around 109 million tonnes. Lastly, the most significant driver of this change is the reduction in landfilling (grey bars), which collapsed by over 50% from 112 million tonnes down to just 49 million tonnes.

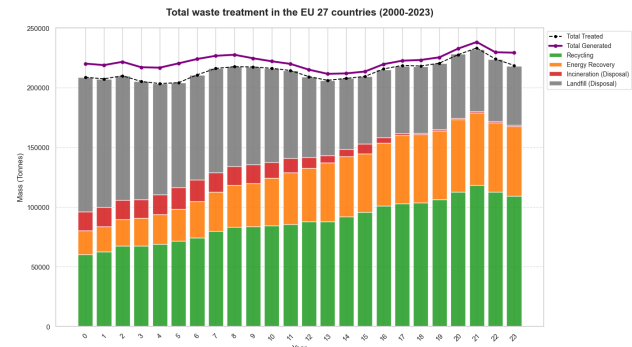


Figure 1: **Structural Shift in Waste Treatment (2000-2023):** The chart visualizes the decline of Landfilling (grey) and the simultaneous rise of Recycling (green) and Energy Recovery (orange), while the total waste volume (black line) remained relatively stable.

In summary, the EU has done massive improvements in waste treatment over the last 23 years and sets a clear trend to sustainable waste treatments. However, in the next section we show that the numbers vary significantly among member states and waste streams.

3.2 How does recycling compare across countries in the EU, and how does it compare for specific types of waste?

We have examined country-level trends to compare how recycling rates, across different waste streams, have evolved across EU member states. First, we compared the individual country's recycling rates with the goals set by the EU. Note, for E-Waste the goals are not as clearly stated.

In Figure 2 we can see that, while over half of the nations have already surpassed the 2025 packaging target of 65%, only one-third (33.3%) have successfully met

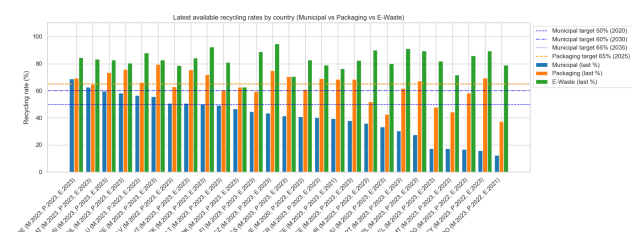


Figure 2: Comparison of National Recycling Performance Against EU Targets (Municipal, Packaging, and E-Waste)

the 2020 municipal target of 50%. There are significant differences among the countries, as seen in the graph and the Kruskal-Wallis test confirms that at least one country is different from all the others for all waste types. Further, as depicted in figure 2, the recycling rates for E-waste are highest for almost all the countries, whereas municipal waste recycling rates are usually the lowest; hence, here we have identified potential for improvement.

Further, those countries showing major up- or down-side developments during the observed period of time have been analyzed in more detail. With so many overlapping factors and policies at play, attributing these shifts to one specific cause is difficult. In general, the major downward trends can be explained by a saturation effect of very advanced recycling nations. Extreme outliers on the up-side (e.g. E-waste recycling development in Malta or Greece) have mostly external factors as a reason instead of incredibly effective policies. For municipal waste, three policies that we have identified have shown very promising results: landfill taxes, residual waste taxes, and deposit return systems.

3.3 Are there characteristics of countries that could lead to increased recycling?

The results of the regression model show that population density exhibits the strongest effects and has a consistently negative impact on municipal and electrical recycling, indicating that denser populations are associated with lower recycling rates in these waste categories. Its effect on packaging recycling is also negative, but less prominent. GDP per capita has a negative and statistically significant effect on packaging recycling, while its effects on municipal and electrical recycling are positive but not statistically significant. Educational attainment shows positive but statistically insignificant effects across all waste categories. Waste expenditure exhibits weak and inconsistent effects overall.

Using PCA and hierarchical clustering for a more exploratory across-country analysis shows that the countries can be split largely into three main clusters. Cluster 1 (orange in Fig. 4) contains countries such as Luxembourg (LU), the Netherlands (NL), Austria (AT) etc. and Cluster 2 (green in Fig. 4) contains countries such as Romania (RO) and Hungary (HR) etc. The across-country methods broadly observe that high-income and countries with high education attainment tend to have higher recycling rates. Population density and waste expenditure show a negative effect on electrical recycling rates. The third cluster contains only Malta (MT) which is an outlier in the PCA as well (see Fig. 3, blue in Fig. 4).

3.4 How well can the development of the amount of recycling be predicted?

Separated by country and waste type, the recycling rate can be predicted well with model based on only past values for up to 5 years. The mean error values over all of the countries can be seen in Table 1. All values are small indicating a good fit.

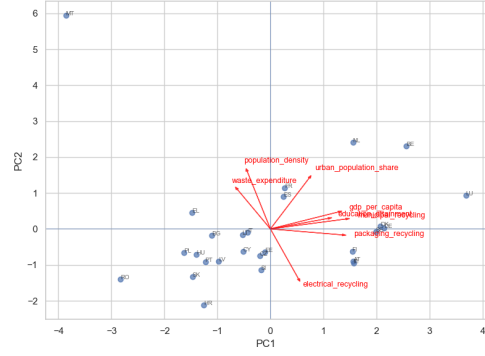


Figure 3: **PCA** on scaled, aggregated data over all years. The first two PCs make up ca. 70% of the cumulative explained variance.

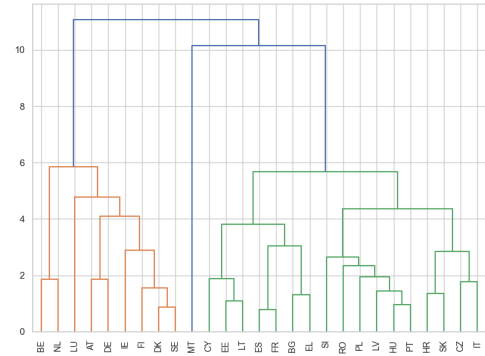


Figure 4: **Hierarchical clustering** on scaled, aggregated data over all years: three emerging clusters can be seen differentiated by the three colors.

	Municipal recycling	Electrical recycling	Packaging recycling
Mean MAE	5.29	4.35	5.48
Mean RMSE	6.09	5.11	6.23

Table 1: Summary of prediction errors across recycling types.

Figure 5 shows one example prediction for Austria for electrical waste. We can see that the prediction is not perfect and drops at the end, even though the true values don't. For the first three years, however, the prediction is great. So, as a conclusion we can say that we can not predict far into the future, but at least for some years.

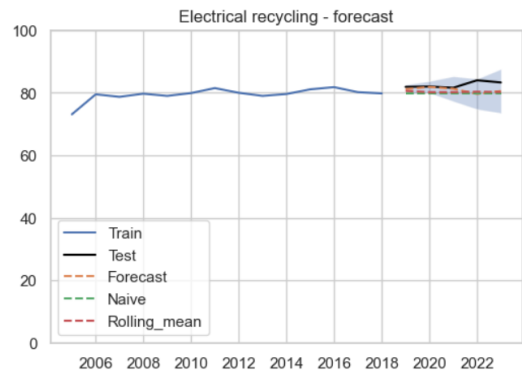


Figure 5: **Prediction** of electrical waste for Austria